Specialist Systems by Industry and Provider

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1 Introduction

This document provides a practical catalogue of specialist systems by industry, with representative providers. It is intended as a reference for enterprise architecture, IT strategy, operating model design, digital transformation, and industry capability mapping.

The catalogue is structured using the following fields:

- **Industry**
- **Specialist Capability**
- **System Type**
- **Purpose**
- **Typical Providers**

This is not an exhaustive market analysis. Provider names are illustrative and represent common or well-known vendors in each category.

2 How to Use This Catalogue

This reference can support several use cases:

- enterprise architecture inventories
- target operating model design
- application portfolio assessments
- industry capability mapping
- vendor landscape overviews
- digital transformation discovery work
- PMO and IT governance documentation

A useful extension is to add fields such as:

- business owner
- technical owner
- deployment model
- integration points
- criticality
- regulatory relevance
- data classification
- lifecycle status

3 Catalogue of Specialist Systems by Industry

Industry	Specialist Capability	System Type	Purpose	Typical Providers
Industrial / Manufacturing	Plant supervision and control	SCADA	Supervisory monitoring and control of industrial processes and equipment	Siemens, Schneider Electric / AVEVA, Rockwell Automation, Inductive Automation
Industrial / Manufacturing	Continuous process control	DCS	Distributed control of continuous and batch industrial processes	Siemens, Emerson, Honeywell, Yokogawa
Industrial / Manufacturing	Manufacturing operations execution	MES / MOM	Manage production execution, quality, scheduling, and shop-floor visibility	Siemens, Rockwell Automation, AVEVA, Dassault Systèmes
Industrial / Manufacturing	Product engineering lifecycle	PLM	Manage product design, engineering changes, and lifecycle data	Siemens, PTC, Dassault Systèmes, Autodesk
Industrial / Manufacturing	Industrial asset maintenance	APM / EAM	Manage reliability, predictive maintenance, and asset performance	IBM Maximo, Hexagon, ABB, GE Digital
Utilities / Energy	Grid operations management	ADMS / EMS / OMS	Operate energy distribution, outages, and grid control	GE Vernova, Schneider Electric, Oracle, Siemens

Industry	Specialist Capability	System Type	Purpose	Typical Providers
Utilities / Water	Customer and billing operations	CIS / Utility Billing	Manage customer records, billing, meter-to-cash, and service orders	Oracle Utilities, SAP, Hansen, Gentrack
Utilities / Water	Meter data operations	MDM	Collect, validate, and process smart meter data	Oracle Utilities, Itron, Siemens, Landis+Gyr
Oil and Gas	Reservoir and production management	E&P / Reservoir Systems	Model reservoirs and manage exploration and production data	SLB, Halliburton, Baker Hughes, Landmark
Oil and Gas	Production operations	Production Management Systems	Monitor wells, production flows, and optimisation	SLB, AVEVA, Baker Hughes, Emerson
Mining	Mine planning	Mine Planning Systems	Plan pit design, extraction, and geological modelling	Hexagon, Micromine, Dassault GEOVIA, Maptek
Mining	Fleet and dispatch	Fleet Management Systems	Control mobile equipment dispatch, haulage, and productivity	Hexagon, Modular Mining, Epiroc, Caterpillar
Healthcare	Clinical patient record	EHR / EMR	Maintain longitudinal electronic health records and clinical workflows	Epic, Oracle Health, MEDITECH, Altera Digital Health
Healthcare	Medical imaging management	PACS / RIS / Enterprise Imaging	Manage radiology workflows and image storage/retrieval	GE HealthCare, Philips, Sectra, Fujifilm

Industry	Specialist Capability	System Type	Purpose	Typical Providers
Healthcare	Laboratory operations	LIS / LIMS	Manage samples, test workflows, and laboratory reporting	Thermo Fisher, LabVantage, Clinisys, Abbott
Healthcare	Hospital operations	PAS / Hospital Information System	Manage admissions, discharges, transfers, scheduling, and administration	Dedalus, Oracle Health, InterSystems, System C
Pharmaceuticals / Life Sciences	Quality and compliance	QMS	Manage deviations, CAPA, training, and regulated quality records	MasterControl, Veeva, Sparta Systems, ETQ
Pharmaceuticals / Life Sciences	Laboratory management	LIMS	Manage scientific lab processes and regulated sample workflows	LabWare, Thermo Fisher, LabVantage, STARLIMS
Pharmaceuticals / Life Sciences	Clinical trial operations	CTMS / eClinical	Manage study planning, monitoring, and clinical trial execution	Medidata, Veeva, Oracle Health Sciences, IQVIA
Pharmaceuticals / Life Sciences	Regulated manufacturing execution	MES	Manage batch production, genealogy, and electronic batch records	Körber Pharma, Siemens, Rockwell Automation, Werum
Banking	Core account processing	Core Banking System	Manage deposits, lending, accounts, and core banking transactions	FIS, Fiserv, Finastra, Temenos, Oracle
Banking	Payments processing	Payments Hub / Payments Engine	Process domestic and international payments and clearing	FIS, ACI Worldwide, Volante Technologies, Temenos

Industry	Specialist Capability	System Type	Purpose	Typical Providers
Banking	Treasury and trading	Treasury / Front-to-Back Systems	Support trading, treasury, settlements, and risk operations	Murex, Calypso, Finastra, FIS
Banking	Loan origination and servicing	Lending Platforms	Manage credit applications, approvals, servicing, and collections	nCino, Temenos, Finastra, Fiserv
Insurance	Policy lifecycle management	Policy Administration System	Manage underwriting, policies, endorsements, and renewals	Guidewire, Duck Creek, Sapiens, TIA Technology
Insurance	Claims handling	Claims Management System	Manage claims intake, workflow, settlement, and reserves	Guidewire, Duck Creek, Sapiens, Origami Risk
Insurance	Billing and collections	Billing System	Manage premium invoicing, collections, and commission handling	Guidewire, Duck Creek, Sapiens, Insurity
Telecoms	Business support	BSS	Support billing, CRM, order management, and revenue operations	Amdocs, Netcracker, Ericsson, Oracle
Telecoms	Network and service operations	OSS	Support provisioning, fault, inventory, and service assurance	Amdocs, Netcracker, Nokia, Ericsson
Telecoms	Mediation and charging	Charging / Mediation Platforms	Process usage records and charging events	Amdocs, Netcracker, Nokia, Ericsson

Industry	Specialist Capability	System Type	Purpose	Typical Providers
Airlines	Airline commercial operations	PSS	Manage reservations, ticketing, inventory, and departure control	Amadeus, Sabre, Navitaire
Airlines	Airline operational control	OCC Systems	Support flight operations, crew coordination, and disruption management	Lufthansa Systems, Sabre, Jeppesen, IBS Software
Aviation Maintenance	Aircraft maintenance and engineering	MRO Systems	Manage maintenance planning, compliance, parts, and records	AMOS, Ramco, IFS, Lufthansa Technik
Airports	Airport operations	AODB / Airport Operations Systems	Manage gates, stands, turnaround, and airport resource allocation	Amadeus, SITA, INFORM, ADB SAFEGATE
Travel Distribution	Travel inventory distribution	GDS	Distribute flights, hotels, and travel bookings	Amadeus, Sabre, Travelport
Rail	Rail traffic control	Traffic Management / Signalling Support	Support dispatching, train movement, and operational control	Siemens Mobility, Hitachi Rail, Alstom, Thales
Rail	Timetabling and rostering	Rail Planning Systems	Manage timetable planning, resources, and crew allocations	Trapeze, Hitachi Rail, Siemens Mobility, IVU

Industry	Specialist Capability	System Type	Purpose	Typical Providers
Maritime / Shipping	Terminal operations	TOS	Manage container yard, vessel planning, and terminal execution	Kaleris Navis, Tideworks, CyberLogitec
Maritime / Shipping	Fleet and voyage operations	Maritime Operations Systems	Manage fleet performance, voyage planning, and compliance	DNV, ABS, Veson Nautical, Kongsberg
Logistics / Warehousing	Warehouse execution	WMS	Manage storage, picking, packing, and warehouse control	Manhattan Associates, Blue Yonder, SAP, Oracle
Logistics / Transport	Transportation planning	TMS	Manage route planning, freight execution, and shipment visibility	Blue Yonder, Manhattan Associates, Oracle, SAP
Logistics / Parcel	Parcel and last-mile operations	Parcel Management Systems	Manage sortation, dispatch, tracking, and delivery operations	Descartes, Metapack, Bringg, FarEye
Retail	Store sales operations	POS	Process in-store sales, promotions, receipts, and cashier workflows	NCR Voyix, Toshiba Global Commerce Solutions, Oracle Retail, Diebold Nixdorf
Retail	Merchandising and replenishment	Retail Merchandising Systems	Manage assortment, pricing, inventory, and replenishment	Oracle Retail, SAP, Blue Yonder, Manhattan Associates

Industry	Specialist Capability	System Type	Purpose	Typical Providers
Retail	Ecommerce commerce platform	Ecommerce Platform	Manage online catalogue, basket, checkout, and fulfilment	Shopify, Adobe Commerce, Salesforce Commerce Cloud, SAP Commerce Cloud
Hospitality	Hotel property operations	PMS	Manage reservations, front desk, room inventory, and guest services	Oracle OPERA, Infor HMS, Mews, Agilysys
Hospitality	Central reservations	CRS	Manage room inventory distribution and hotel booking channels	Oracle Hospitality, SHR, Sabre Hospitality, Amadeus
Restaurants / Food Service	Restaurant operations	Restaurant POS / Back Office	Manage table service, orders, tills, kitchen, stock, and labour	Toast, NCR Voyix, Oracle MICROS, Square
Construction / Engineering	Built asset design coordination	BIM / CDE	Manage building models, drawings, and common data environments	Autodesk, Bentley, Trimble, Nemetschek
Construction / Capital Projects	Project controls	CPM / Project Controls Systems	Manage schedules, cost, progress, risk, and earned value	Oracle Primavera, Hexagon EcoSys, Deltek
Construction / Engineering	Field operations and site management	Construction Management Systems	Manage site activities, forms, quality, and punch lists	Procore, Autodesk Construction Cloud, Oracle Aconex, Trimble

Industry	Specialist Capability	System Type	Purpose	Typical Providers
Real Estate	Property administration	Property Management System	Manage leases, tenants, rent, maintenance, and real estate operations	Yardi, MRI Software, RealPage
Education	Student administration	SIS	Manage student records, admissions, enrolment, grades, and timetables	Ellucian, PowerSchool, Tribal, Unit4
Education	Learning delivery	LMS / VLE	Deliver courses, assignments, and online learning experiences	Moodle, Canvas, Blackboard, D2L
Government / Public Safety	Emergency dispatch	CAD	Support call handling, dispatch, and incident coordination	Motorola Solutions, Hexagon, CentralSquare, Tyler Technologies
Government / Public Safety	Law enforcement records	RMS	Manage incident reports, case records, and operational policing data	Motorola Solutions, Hexagon, CentralSquare, Mark43
Government / Tax	Revenue collection and compliance	Tax and Revenue Systems	Manage tax records, filings, assessments, and collections	FAST Enterprises, SAP, Oracle
Justice / Courts	Court administration	Case Management Systems	Manage hearings, filings, case progression, and judicial workflow	Tyler Technologies, Thomson Reuters, Journal Technologies
Media / Broadcast	Media content operations	MAM	Manage media assets, metadata, storage, and production workflows	Avid, Dalet, Vizrt, Evertz

Industry	Specialist Capability	System Type	Purpose	Typical Providers
Media / Broadcast	News production	Newsroom Systems	Support story planning, rundowns, and newsroom collaboration	ENPS, Avid, Dalet, Octopus Newsroom
Agriculture	Farm operations	Farm Management Systems	Manage crop planning, field operations, and input usage	John Deere, Trimble Agriculture, Climate FieldView, AGCO
Agriculture	Precision agriculture	Precision Ag Platforms	Optimise seeding, spraying, yield, and machine telemetry	John Deere, Trimble, Raven, Climate FieldView
Legal	Legal operations and matter handling	Practice Management / Matter Management	Manage cases, billing, time recording, and legal workflows	Clio, iManage, Thomson Reuters, Aderant
Professional Services	Engagement and resource operations	PSA	Manage projects, utilisation, time, billing, and consulting delivery	Deltek, Kantata, Certinia, Mavenlink
Energy Trading	Commodity and market operations	ETRM / CTRM	Manage energy trades, risk, scheduling, and settlements	ION, Allegro, Eka Software, FIS
Automotive	Dealer and aftersales operations	Dealer Management System	Manage sales, service, parts, and dealership finance	CDK Global, Reynolds and Reynolds, Keyloop
Automotive	Vehicle lifecycle and engineering	Automotive PLM / ALM	Support product engineering, compliance, and design lifecycle	Siemens, PTC, Dassault Systèmes, Autodesk

4 Cross-Industry Observations

Although these systems are industry-specific, they usually fall into a smaller number of architectural categories.

4.1 Operational Control Systems

These directly control or monitor operations, often in real time.

Examples include:

- SCADA
- DCS
- EMS
- ADMS
- OSS
- fleet dispatch systems

4.2 Core Transaction Systems

These execute the primary business transactions of an industry.

Examples include:

- core banking
- policy administration
- PSS
- PMS
- SIS
- POS

4.3 Record and Compliance Systems

These provide legally or operationally important records.

Examples include:

- EHR
- PACS
- LIS
- CTMS
- RMS
- case management systems

4.4 Planning and Execution Systems

These support operational planning, workflow, and optimisation.

Examples include:

- WMS
- TMS
- MES
- project controls systems
- mine planning systems
- farm management systems

5 Suggested Enterprise Architecture Extension

You can extend the catalogue with a more complete inventory model such as the following.

Field	Description
Industry	Industry domain or sector
Specialist Capability	Business capability supported by the system
System Type	Recognised application category
Business Purpose	Main operational or strategic purpose
Provider	Vendor or service provider
Product Name	Specific implementation or product used
Deployment Model	On-premise, cloud, SaaS, hybrid
Business Owner	Accountable business function
Technical Owner	IT or platform owner
Criticality	Mission critical, high, medium, low
Integration Points	Upstream and downstream systems
Data Sensitivity	Public, internal, confidential, regulated
Regulatory Relevance	Applicable regulations or standards
Lifecycle Status	Strategic, active, legacy, retiring
Notes	Additional remarks

6 Example Application Inventory Entry

Below is an example of how a specialist system can be represented in an enterprise architecture repository.

Field	Example Value
Industry	Industrial / Manufacturing
Specialist Capability	Plant supervision and control
System Type	SCADA
Business Purpose	Monitor and control plant assets in real time
Provider	Siemens
Product Name	WinCC
Deployment Model	Hybrid
Business Owner	Head of Manufacturing Operations
Technical Owner	Industrial Platforms Manager
Criticality	Mission critical
Integration Points	PLCs, historian, MES, EAM, reporting
Data Sensitivity	Confidential
Regulatory Relevance	Health and safety, cyber security, operational resilience
Lifecycle Status	Strategic
Notes	Requires strong network segmentation and OT security controls

7 Architecture Notes

7.1 Specialist Systems Are Usually Not Standalone

Most specialist systems are heavily integrated with broader enterprise platforms such as:

- ERP
- CRM
- identity and access management
- data warehouse / data lake
- integration middleware
- reporting and analytics platforms
- document management systems
- monitoring and cyber security tooling

7.2 Industry Systems Often Drive Core Value Creation

In many industries, the specialist system is the operational core. For example:

- a bank depends on core banking
- an airline depends on its PSS

- a hospital depends on EHR and PACS
- a utility depends on ADMS/OMS/CIS
- a factory depends on SCADA, DCS, and MES

7.3 Governance Considerations

When cataloguing specialist systems, it is useful to assess them through several lenses:

- business criticality
- resilience requirements
- information security
- compliance and regulation
- vendor lock-in risk
- integration complexity
- data ownership
- lifecycle and technical debt

8 Practical Use in IT Strategy

This catalogue can support strategic analysis such as:

8.1 Application Portfolio Review

Identify:

- duplicate systems
- unsupported legacy systems
- strategic systems
- systems needing modernisation
- cyber and resilience priorities

8.2 Capability Mapping

Map applications to business capabilities and determine:

- which capabilities are well supported
- which depend on manual processes
- where transformation investment is needed

8.3 Target Architecture Design

Use the catalogue to define:

- strategic platforms
- integration principles
- data ownership
- coexistence between legacy and modern platforms

9 Conclusion

Specialist systems vary by industry, but the design logic is similar across sectors. Every industry tends to rely on a combination of:

- operational control systems
- core transaction systems
- record and compliance systems
- planning and execution systems

A structured catalogue helps organisations understand where these systems sit in the enterprise landscape, how they support business capabilities, and where they create operational dependency, risk, and strategic value.

10 References

Representative provider and product categories can be validated through official vendor sites and product documentation from vendors such as Siemens, Schneider Electric / AVEVA, Rockwell Automation, Epic, Oracle Health, Temenos, Guidewire, Amdocs, Amadeus, Manhattan Associates, Oracle Hospitality, Autodesk, and others.

For formal publication, you may wish to replace the general reference note above with a full bibliography of official vendor product pages and analyst references.